

iKwath

Pod-Based Smart Kwatha (Kadha) Maker

SIH 2026 | Problem ID 26048 | Consolidated Technical & Design Proposal

1. Executive Summary

iKwath is a reusable countertop machine paired with single-dose powder pods that prepares a fresh, AFI/API-standardized Kwatha (Ayurvedic decoction) from coarse powder (yavakut churna) on demand. It targets a preparation time of 20–25 minutes per dose, a 2–2.5x reduction from the 45–60 minutes typically required by classical AFI-method preparation, achieved purely through evaporation-surface and vapour-path engineering rather than higher temperature — so speed is gained without altering the underlying reduction chemistry.

The system replaces the most variable step in classical preparation — judging the reduction endpoint by experience — with a measurement-driven control loop. A load cell and temperature probe continuously track the decoction as it reduces, and the machine's core algorithm, **Adaptive dm/dt Endpoint Detection**, stops the process exactly when the validated endpoint for that formulation is reached. Each pod is pre-measured to the correct AFI/API ratio (commonly 1 part powder : 16 parts water, reduced to ¼ volume) and doubles as the filter medium, so the user only inserts a pod and adds water.

Equivalence to the classical reference method is treated as something to be proven, not assumed: every formulation is validated against a reference-vs-machine comparison (yield, markers, pH, colour, odour, taste, and volatile-compound retention) before it is accepted into the recipe library. The first prototype is deliberately scoped to one formulation and eight core hardware elements, with a phased go-to-market path from Ayurvedic clinics, to pharma QC labs, to eventual consumer retail.

2. Problem Definition and Design Requirements

Problem statement: Design a pod-based smart Kwatha maker that prepares a fresh, AFI/API-standardized decoction from coarse powder (yavakut churna) on demand, in the shortest practical time, without altering the decoction's quality or yield.

Classical Kwatha preparation is manual, slow, and operator-dependent. The user must measure powder and water in the correct AFI/API ratio, sustain heat for an extended period, judge the reduction endpoint by experience, filter the spent powder, and clean the vessel. Because judgement replaces measurement at the most critical step — the endpoint — batch-to-batch quality varies, and the long preparation time discourages daily use.

Benchmark and design targets:

Requirement	Target / Constraint
Preparation time	20–25 minutes per dose, vs. 45–60 minutes for classical AFI-method preparation (a 2–2.5x reduction)
Dosing standard	Full compliance with the AFI/API pharmacopoeial ratio for each formulation (commonly 1 : 16, reduced to ¼ volume)
Quality & yield	Must not be altered to gain speed; speed comes from evaporation/vapour engineering, not higher temperature

Requirement	Target / Constraint
Volatile retention	Aromatic/volatile fractions relevant to taste and therapeutic character must be substantially retained, not vented away
Consistency	Batch-to-batch variation must be minimised by replacing operator judgement with sensor-driven control
Hygiene	Must support safe multi-patient / clinical reuse without cross-contamination
Cost	Must be buildable from a low-cost, off-the-shelf bill of materials
Regulatory realism	Presented as an engineering prototype with an equivalence-testing protocol, not a finished regulatory approval

3. Proposed Solution and Features

iKwath is a reusable countertop machine paired with single-dose powder pods. Each pod holds the coarse powder for one formulation, pre-measured to the correct AFI/API ratio, and doubles as the filter medium during extraction. The machine stores a per-formulation recipe — water quantity, soak time, temperature, extraction, reduction, validated endpoint, and filtration — and does not rely on a fixed timer; instead it tracks real-time weight and temperature to decide when the decoction is actually ready.

Key features:

- Single-dose pods pre-formulated to the classical AFI/API ratio, calibrated against the pharmacopoeial standard before being added to the recipe library.
- Sensor-driven, measurement-based endpoint detection instead of a fixed timer (Section 15).
- Evaporation-surface and vapour-path engineering that speeds reduction without raising temperature beyond the validated band (Section 5).
- A vapour outlet with partial condensation/return that protects volatile/aromatic compounds relevant to taste.
- The pod itself acts as the filter medium, removing a separate filtration step for the user.
- A live on-device display that exposes the controller's real-time reasoning (Section 8).
- An autoclavable, food-grade chamber suited to daily clinical-cycle counts (Section 14).
- A structured reference-vs-machine equivalence protocol before any formulation is trusted (Section 6).

4. How the Machine Works

In one line: the machine watches the process continuously, instead of asking the user to watch it. The operating sequence is as follows.

Step	Action	What is controlled
1	Insert pod + add water	Correct powder mass + AFI-ratio water volume; pod ID auto-read
2	Soak	Powder evenly wetted before heat is applied
3	Controlled extraction	Mild heat + gentle stirring for uniform mass transfer
4	Faster reduction	Enlarged evaporation surface + managed vapour path (not just higher heat)
5	Endpoint check	Load cell (mass) + temperature sensor jointly confirm the validated endpoint
6	Filter + dispense	Pod acts as filter; ~100 mL fresh warm Kwatha delivered

Step	Action	What is controlled
7	Rinse / clean	Chamber + filter path cleaned; ready for the next pod

What happens inside each stage:

- **Pod:** the machine reads the pod ID and loads the matching validated recipe.
- **Water:** AFI/API-ratio water is added for the selected formulation (worked example: ~400 mL).
- **Soaking:** powder absorbs water before heat is applied, improving extraction uniformity.
- **Heating:** a PID-controlled heater holds the formulation-specific band (worked example: ~85–90°C, confirmed per formulation against the reference method).
- **Mixing:** gentle stirring improves powder–water contact and extraction efficiency.
- **Reduction:** water is removed by controlled evaporation using larger effective surface area and managed vapour removal, not simply a hotter heater.
- **Smart endpoint:** the load cell tracks falling mass in real time; the controller combines mass and temperature trends to detect the validated endpoint (Section 15).
- **Filtering & dispensing:** the pod separates spent powder from the liquid, and ~100 mL of fresh warm Kwatha is delivered.
- **Cleaning:** the chamber and filter path are rinsed / are autoclavable, readying the unit for the next dose.

5. Technical Approach: How the Machine Makes It Faster

The problem statement explicitly requires speed without altering quality or yield. iKwath does not raise temperature to save time; it raises the rate at which water can leave the liquid while holding the formulation's validated temperature band.

Method	Purpose	Effect on Quality
Larger effective evaporation surface	More liquid surface exposed for evaporation	Neutral – geometry only
Controlled vapour removal + partial return	Prevents vapour accumulation; limits volatile loss	Protects aroma/marker retention
Gentle stirring / recirculation	Improves heat and mass transfer	More uniform extraction
Mass-loss (dm/dt) monitoring	Confirms reduction is on the expected curve	Enables early anomaly detection
Adaptive control within validated limits	Fine-tunes rate inside approved bounds only	Never exceeds validated temperature/time bounds

The best combination of these methods is selected experimentally, comparing cycle time against the reference-sample equivalence tests (Section 6) — a faster cycle is only accepted into the recipe library if it still passes equivalence.

6. Proving “Same as AFI” — Equivalence, Not a Claim

“Same taste as AFI” is treated as a testable equivalence target, not a marketing line. For each selected formulation, two samples are prepared and compared directly: a Reference sample made by the accepted classical/AFI method, and a Machine sample made by iKwath using the same formulation's recipe.

Comparison parameters include final volume, extractive yield / total solids, pH (where relevant), colour, odour, organoleptic (taste) evaluation, and selected chemical marker compounds or a chromatographic (TLC/HPTLC) profile. Volatile/aromatic marker retention is included explicitly, directly answering the concern that faster evaporation could strip aroma-relevant compounds. Machine settings are accepted into the recipe library only when the machine sample falls within predefined limits of the reference sample.

Illustrative reporting format (values are placeholders, not results):

Parameter	Reference	Machine (illustrative)
Extractive yield	100%	98.7%
Marker compound A	100%	99.2%
Final volume	100 mL	100 mL
Preparation time	60 min	35 min

These four numbers are placeholders that show the reporting format only; real lab results from the chosen formulation must replace them before submission or demonstration. The project's approach is: develop → compare against the selected reference preparation → measure → tune → repeat → demonstrate equivalence within defined limits — not to assume an identical AFI taste automatically.

7. Live Display — Making the Control Logic Visible

During a live demonstration, the biggest risk is that observers see a closed box that simply makes liquid appear. The on-device display is deliberately used to expose the controller's live reasoning, so a viewer is watching the same weight-and-temperature decision the machine is making in real time.

```
FORMULATION: XXXXX
Water       : 400 mL
Temperature : 88°C
Current mass : 165 g
Target mass  : 100 g
Stage: REDUCTION
Progress: ██████████ 82%
Estimated remaining: XX min
```

This turns a demonstration from “trust the machine” into “watch the machine control the process.” An observer can see the formulation, live mass vs. target, current stage, and a progress/ETA readout updating in real time — a direct, visible proof of the Adaptive dm/dt endpoint logic described in Section 12.

8. Proposed System Architecture and Flow Diagram

The architecture is organised around a central controller that coordinates sensing, thermal, mixing, and output subsystems using a recipe state machine.

Subsystem	Component	Function
Control	ESP32 (or similar MCU)	Runs the recipe state machine and logs the weight/temperature trace
Sensing	Load cell (chamber base)	Continuous mass tracking to compute the dm/dt reduction rate
Sensing	Food-grade temperature probe	Tracks heating, extraction, and reduction temperature
Sensing	Pod ID (RFID/QR/contact code)	Auto-loads the correct recipe per formulation
Thermal	Electric heater, PID-controlled	Maintains the formulation-specific temperature band
Mixing	Low-speed stirrer	Improves heat and mass transfer without excess turbulence
Vapour path	Controlled vapour outlet	Speeds evaporation while limiting volatile-oil loss
Output	Filter + valve/pump	Separates spent powder and dispenses the ~100 mL dose
Chamber	Removable stainless vessel	Food-grade, autoclavable for clinical reuse
UI	Small LCD/OLED + buttons	Formulation selection, progress, and endpoint confirmation

iKwath — End-to-End Control & Process Flow

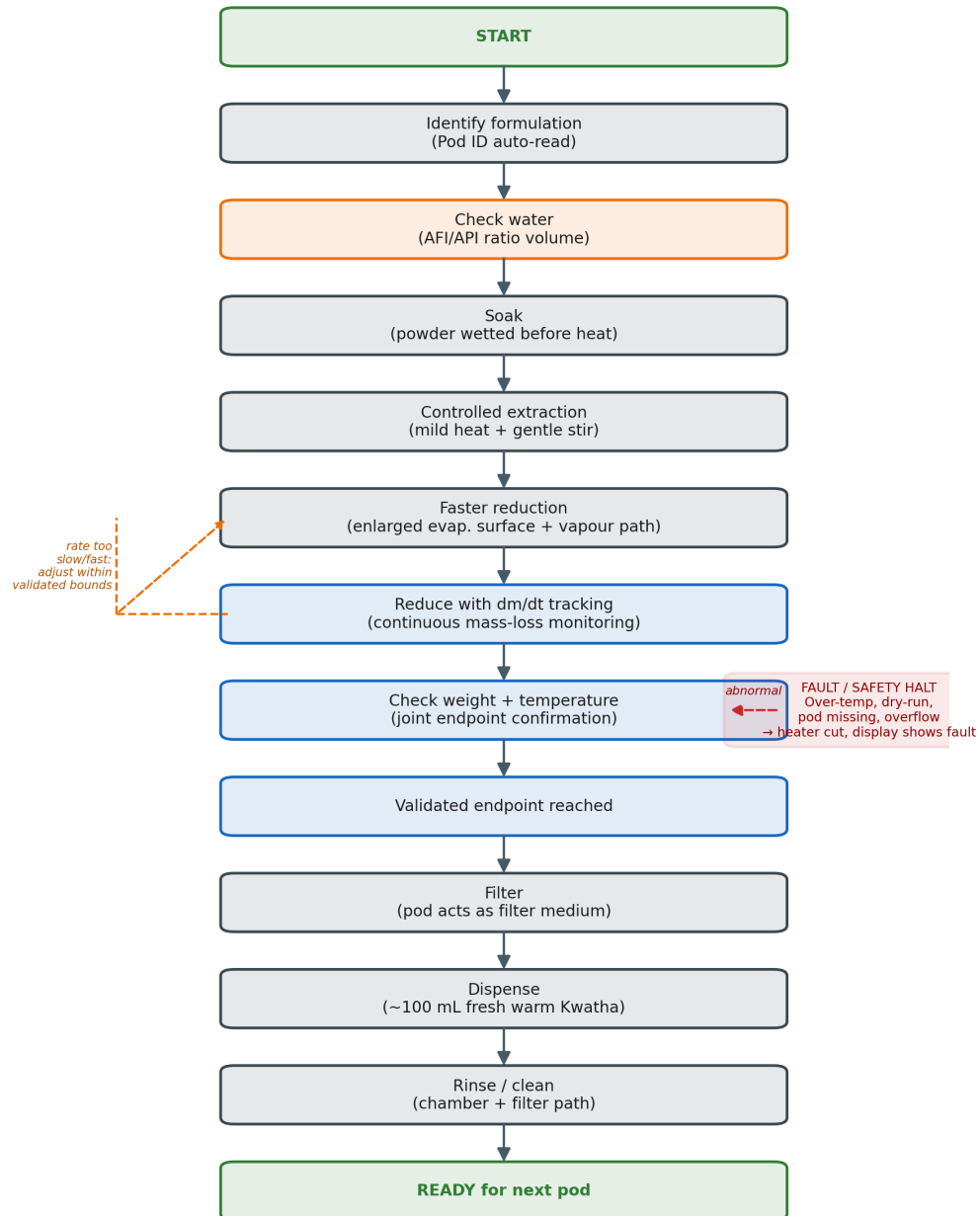


Figure 1. End-to-end process and control flow, from pod insertion through dispensing to the ready state, including the two safety/adaptive branches (dm/dt-triggered rate adjustment, and the fault-halt interlock).

9. End-to-End Working Protocol

The complete control sequence, as executed by the recipe state machine on every cycle:

START → Identify formulation (pod ID) → Check water → Soak → Heat → Extract → Reduce (dm/dt tracked) → Check weight + temperature → Validated endpoint reached → Filter → Dispense → Rinse → **READY**

Safety interlocks (mandatory in the first prototype, not deferred):

- If temperature exceeds the validated band, heater output is reduced or cut.
- If weight/water level is abnormal (e.g. dry-run, pod missing, overflow), the cycle halts and the display shows the fault.

From the user's perspective the entire cycle is a single control-panel operation: insert the pod, add the specified water, select the formulation if not auto-detected, and let the machine run through soak, heat, extraction, reduction, endpoint detection, filtering, and dispensing unattended, finishing with an automatic rinse that prepares the unit for the next pod.

10. Technical Strategy for Minimising Preparation Time

Because temperature is fixed by the validated band for each formulation, time reduction is pursued entirely through physical and control-system engineering:

- **Evaporation-surface engineering:** an enlarged effective liquid surface area increases the evaporation rate at a constant, validated temperature.
- **Vapour-path management:** a controlled outlet with partial condensation/return accelerates net water removal while limiting loss of volatile/aromatic fractions.
- **Mass-transfer improvement:** gentle stirring/recirculation improves heat and mass transfer, shortening the time needed to reach uniform extraction.
- **Continuous dm/dt monitoring:** real-time mass-loss tracking confirms the reduction is following the expected curve, allowing early detection of a stalled or overshooting process instead of a fixed, conservative timer.
- **Adaptive control within validated bounds:** the rate is fine-tuned inside pre-approved limits only, so the process is never sped up by exceeding a validated temperature or time bound.

Each candidate combination of these methods is tested against the reference-sample equivalence protocol (Section 6); a faster cycle time is adopted only if the resulting decoction still passes equivalence testing, which is what allows iKwath to target 20–25 minutes against a 45–60 minute classical benchmark without treating speed and quality as a trade-off.

11. Qualitative and Quantitative Quality Presentation

Quality is validated on two tracks: measurable, quantitative parameters that can be logged and charted, and qualitative organoleptic parameters that require sensory comparison.

Validation parameters measured:

Test	What Is Measured
Preparation time	Reference method vs. iKwath, same formulation
Water quantity	Starting water accuracy against the AFI ratio
Temperature profile	Actual temperature-vs-time trace vs. the validated band
Reduction	Final mass/volume vs. target
Extractive yield	Amount of extracted material vs. reference
Chemical profile	Selected formulation marker compounds
Volatile retention	Aroma-relevant marker loss, machine vs. reference
Taste / appearance	Organoleptic comparison, blinded where feasible
Repeatability	Variance across multiple machine batches
Cleaning	Time and ease of cleaning / turnaround between doses

Quantitative results (yield, marker-compound retention, final volume, preparation time) are presented as a direct reference-vs-machine table, as shown in Section 6, rather than as a verbal claim. Qualitative results (colour, odour, taste) are captured through blinded organoleptic comparison where feasible, so subjective impressions are gathered under the same controlled conditions as the objective measurements. The first prototype is validated on 1–2 selected Kwatha formulations, with additional formulations added to the recipe library only after passing this same equivalence protocol.

12. Control System, Sensors and Hardware

Low-cost prototype bill of materials:

Part	Choice for First Prototype
Controller	ESP32 / similar microcontroller
Temperature sensor	Food/process-rated temperature sensor
Weight sensor	Load cell + HX711
Heater	Electric heater with PID temperature control
Mixing	Small low-speed motor / stirrer
Chamber	Removable stainless-steel vessel (autoclavable)
Pod	Single-dose coarse-powder filter pod
Vapour path	Controlled outlet with partial condensation/return
Outlet	Filter + small pump/valve
Display	Small LCD/OLED

The first prototype is deliberately kept to exactly these eight functional elements — load cell, temperature sensor, heater, stirrer, chamber, pod/filter, controller, and display — and nothing more. AI, cloud connectivity, camera-based sensing, and an app ecosystem are deliberately excluded from v1: none of them are needed to prove the core claim of sensor-driven, formulation-specific reduction control, and adding them would dilute build time without adding demo-relevant credibility. They remain valid future extensions, not v1 scope.

Scope discipline: the first prototype targets a single, deliberately chosen Kwatha formulation, so the goal for evaluation is one clean, fully-proven result — “this machine reliably prepares this one formulation faster and more consistently than the classical method, with demonstrated equivalence” — rather than several partially-proven ones. Multi-formulation support is framed as a later platform capability.

13. Pod Design, Hygiene and Safety

Each single-dose pod holds coarse powder pre-measured to the correct AFI/API ratio for one formulation, and carries an ID (RFID/QR/contact code) that lets the machine auto-load the matching validated recipe. The pod also functions as the filter medium during extraction, so spent powder never contacts the machine directly.

Hygiene and multi-patient use:

For clinic and hospital deployment, cross-contamination between successive patients is a real concern. iKwath addresses this with:

- Single-use pods, so no powder ever contacts the machine directly.
- An autoclavable stainless chamber and filter path rated for daily clinical-cycle counts.
- A rinse cycle enforced by the controller before the next pod can be accepted.
- An optional wipeable/sealed housing around all sensor penetrations.

Safety interlocks:

- Over-temperature protection: if temperature exceeds the validated band, heater output is reduced or cut.
- Dry-run / abnormal-level protection: if weight or water level is abnormal (dry-run, missing pod, overflow), the cycle halts and the display shows the fault.

Both interlocks are treated as mandatory requirements for the first prototype rather than features to be deferred to a later version.

14. Core Algorithm: Adaptive dm/dt Endpoint Detection

The endpoint-detection method, named **Adaptive dm/dt Endpoint Detection**, is the project's core novel contribution and its single most demo-able piece of technical depth. Temperature indicates how the liquid is being heated; mass indicates how much liquid remains. The controller does not run on an instruction like “heat for 30 minutes.” It runs on: “continue reduction until the measured process reaches the validated endpoint.”

Illustrative mass trace through a cycle:

Time (illustrative)	Measured Mass	Stage
t0	400 g	Start — soak / heat-up
t1	350 g	Extraction, early evaporation
t2	250 g	Active reduction
t3	180 g	Active reduction, dm/dt tracked
t4	120 g	Approaching endpoint window
t5	100 g	Validated endpoint — stop, filter, dispense

The controller recognizes this curve in real time and stops exactly at the validated endpoint for that formulation, rather than after a fixed duration. It also computes dm/dt (the rate of mass loss) to detect whether evaporation is running too slow (process stalled, possible heater fault) or too fast (risk of overshoot / thermal damage to markers), and adjusts within validated bounds accordingly. This is what makes the endpoint formulation-specific and measurement-driven rather than time-driven.

15. Final Proposal in One Sentence

“A low-cost, pod-based Kwatha machine that prepares an AFI/API-standardized dose in 20–25 minutes using Adaptive dm/dt weight-and-temperature feedback to control reduction, with equivalence to the classical reference method proven — not assumed — before any formulation is added to its recipe library.”

Supporting summary of what makes the solution strong:

Requirement	iKwath's Answer
Real problem	Removes manual judgement from the highest-variance step: the endpoint
Standardization	Pod recipes calibrated to the AFI/API ratio per formulation
Speed, without quality loss	Evaporation-engineering + dm/dt control, not higher heat
Low cost	Off-the-shelf ESP32 / load cell / PID heater bill of materials
Consistency	Recipe-based, sensor-verified process control
Quality proof	Structured reference-vs-machine equivalence protocol
Clinical readiness	Single-use pods + autoclavable chamber
Path to market	Phased clinic → QC lab → consumer rollout
Scalability	New formulations added as validated recipes, not new hardware

This document is a project design proposal. Formulation specifications, food-contact safety, laboratory testing, and any regulatory or therapeutic claims must be verified with appropriate Ayurveda/pharmacopoeia and technical experts before any real-world deployment.

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